SentinelAI

AI-Powered Multi-Modal Digital Threat Detection & Digital Authenticity Analysis Platform

Problem Statement and Topic Selection

Software Project Design and Development
CSE 416

Student 1 (Team Leader)	**Reyazul Islam (ID: 0432320005101146)**
Student 2 (Team Member)	**Sadia Jannat (ID: 0432320005101108)**
Student 3 (Team Member)	**Fatema Akter Nishi (ID: 0432320005101121)**
Department	**Computer Science and Engineering**
Semester	**Autumn 2026**
Course Instructor	**Dr. Mahfida Amjad Dipa, Assistant Professor , Dept. of CSE**
Date of Submission	**1st September 2026**

Problem Statement

The rapid advancement of Generative AI has significantly increased the ability to create convincing **deepfake images, videos, voices, forged documents, phishing websites, scam messages, and malicious files**. These technologies can be exploited for identity impersonation, financial fraud, misinformation, blackmail, and the creation of fake official or government-related documents.

The scale of the broader cybercrime problem is already substantial. According to the FBI Internet Crime Complaint Center (IC3), reported cybercrime losses in the United States reached **$16.6 billion in 2024**, with phishing/spoofing, business email compromise, government impersonation, and other fraud contributing significantly to reported losses. The FBI has also specifically warned that criminals are using Generative AI to make fraudulent content more convincing and easier to produce.

The threat is becoming increasingly interconnected. Recent FBI warnings describe scams involving **AI-generated videos impersonating government officials combined with spoofed government websites**, demonstrating how deepfakes and conventional phishing can be used together in a single attack. Similarly, malicious actors have used AI-generated voice messages and phishing links while impersonating senior officials.

However, detection is often fragmented across different specialized tools. A user may need one service to analyze a deepfake, another to inspect a suspicious URL, another to check a file, and additional tools to investigate a potentially forged document.

Therefore, there is a need for a **unified, AI-powered web platform** that can analyze multiple forms of suspicious digital content and provide a single, understandable assessment of authenticity and cybersecurity risk.

Topic Selection

SentinelAI — AI-Powered Multi-Modal Digital Threat Detection & Digital Authenticity Analysis Platform

SentinelAI is selected as a software development project to address the growing convergence of **AI-generated content, digital forgery, online fraud, and cybersecurity threats**.

The proposed platform will allow users to submit suspicious **images, videos, audio, documents, URLs, messages, and files** through a single web interface. Specialized AI/ML detection modules will analyze each threat category, while a unified risk engine will correlate the detected evidence.

The platform will additionally integrate a **local LLM through Ollama using Llama 3.1 / Qwen3** to interpret technical findings, explain why content is considered suspicious, and provide actionable recommendations.

The topic was selected because it combines **Artificial Intelligence, Cybersecurity, Digital Forensics, Machine Learning, and Web Application Development** into one practical system. More importantly, SentinelAI addresses a real-world problem where a single incident may involve multiple attack vectors—for example, an AI-generated impersonation video directing a victim toward a fake government website or a malicious document.

The project's core objective is therefore not simply to detect whether content is "fake," but to provide users with a **unified digital threat assessment: What is suspicious, why it is suspicious, how severe the risk is, and what the user should do next.**